Physical Chemistry CUME Oct. 28th, 2022

Energy, Heat, and Work

Water boiling at 100 ^{o}C is pushing a piston out by 10 L against 1 atm external pressure. (The initial volume of the gas phase is 0 and the final volume is 10 L) (Assume everything happened in a quasistatic process with no temperature or pressure differential.)

a. How many grams of water become vapor(steam) in the boiling process.

b. What is the work done by the boiling water?

c. What is the heat absorbed by the water? (total heat not heat per mole)

d. What is the change in internal energy for the water? (report total internal energy change, not per mole)

Quantities you may or may not need: C_p (liquid water @ 100 C) = 4.1813 J/(gK), C_p (steam @ 100 C) = 2.080 J/(gK), heat of vaporization (water, 100 C)=2.259 kJ/g, Molecular Mass of water 18.02 g/mol. Ideal Gas Constant 8.314 J/(molK). 1 atm=101325 Pa.